JAMES MCTIGHE

Software Engineering Intern

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Indianapolis, IN

Software Engineer with 6+ years of total technical experience across full-stack development and analytical R&D, specializing in Python, JavaScript/React, and scalable microservice architectures. Proven track record of designing high-performance RESTful APIs, optimizing relational databases (PostgreSQL), and deploying containerized applications via Docker and GitHub Actions. Adept at cross-functional collaboration, code reviews, and maintaining rigorous testing and code quality standards in Agile environments

Work Experience

Software Engineering Intern

Nov 2025 - 2026

OPENLANE | Indianapolis, IN, USA

- Engineered scalable web applications using React and Next.js, optimizing backend server-side rendering to handle high-volume, diverse datasets.
- Implemented CI/CD pipelines to automate software build, test, and deployment processes
- Developed automated features and integrated APIs across business units to ensure software scalability.
- Assisted in deploying and managing applications using Kubernetes, enhancing system reliability and scalability
- Implemented Redis caching strategies to improve application performance and reduce database load
- Assisted in the implementation and testing of NoSQL database structures for various software projects

QC Analyst

Dec 2024 - Oct 2025

Orano Med | Indianapolis, IN

- Developed automation tools using Python to parse complex instrument reports (e.g., ICP-MS, HPLC), converting raw unstructured data into AI-ready formats for downstream R&D analysis and accelerating assay reporting workflows.
- Leveraged SAS and complex SQL queries to identify trends in biochemical assays, supporting early-stage discovery and process optimization.
- Collaborated with lead scientists and project leaders to design custom data extraction solutions that supported feasibility assessments and R&D testing of Lead-212 radiopharmaceutical production.
- Assisted in the migration of applications to Google Cloud, enhancing system performance and reliability.
- Led equipment qualification planning and coordinated cross-functional activities to meet accelerated timelines while maintaining documentation standards.

Senior Chemist

Dec 2021 - Nov 2024

Cardinal Health | Indianapolis, IN

- Utilized SAS (Base, Macro, STAT) to develop CDISC-compliant ADaM datasets, transforming raw biochemical assay results into analysis-ready formats to inform vaccine candidate selection.
- Created a Flask-based web application to modernize site-wide equipment and material tracking, reducing manual lookup time and improving traceability.
- Participated in the full software development lifecycle, from design to testing and deployment
- Led and supported analytical method validation activities (accuracy, precision, linearity, specificity, robustness) for chromatographic assays and supported troubleshooting to improve assay performance.
- Authored and reviewed validation protocols and reports, ensuring alignment with regulatory expectations and internal SOPs.

Data Analyst
Feb 2021 - Nov 2021

Emergent BioSolutions | Indianapolis, IN

- Engineered standardized data collection protocols and ETL workflows to ensure manufacturing datasets were AI- ready for downstream predictive modeling and R&D analysis.
- Modernized legacy data entry workflows by transitioning manual records into LIMS-integrated relational databases, improving data retrieval speed and ensuring codebase sustainability for site-wide quality control.
- Assisted in implementing Devops tools and practices to improve team collaboration and efficiency
- Leveraged Python-based analysis and complex SQL queries to identify trends in biochemical assays, supporting hypothesis generation for early-stage discovery.

Analytical Chemist
May 2019 - Jan 2021

Midwest Compliance Labs | Terre Haute, IN

- Conducted analysis of pharmaceutical stability samples utilizing chromatography methods and instruments.
- Assisted senior scientists in developing and validating LC-MS methods, optimizing sample preparation and instrument parameters to improve throughput and reproducibility of metabolomics assays.
- Performed routine maintenance and troubleshooting of HPLC and GC units to ensure continuous operation and data integrity.

Projects

Reaction Kinetics Web Application (Merck)
Jan 2026 - Present

- Coordinated with Merck subject matter experts to develop a web application to analyze reaction kinetics information.
- Processed Chemstation HPLC reports of different reaction conditions to evaluate how conditions impact reaction rate and inform experimental optimization.

RNA Host Identification
Jan 2025 - May 2025

- Architected an AI Pipeline: Developed a hybrid Convolutional Neural Network (CNN) and Transformer architecture to predict viral mutation and human host infection potential.
- Rigorous Model Evaluation: Achieved 97% accuracy on hold-out test sets. Conducted error analysis to ensure model robustness and reliability in biological contexts.

Natural Disaster Assessment
Jan 2025 - May 2025

- Computer Vision Classification: Engineered a classification model using Scikit-Learn to evaluate both the type and severity of natural disasters from image data.
- Statistical Validation: Implemented cross-fold validation and hyperparameter tuning to ensure model generalizability, achieving 96% accuracy against test cases.

Molecular Property Predictor
Aug 2024 - Dec 2024

- Graph Representation Learning: Created a molecular graph construction pipeline using NetworkX, representing atoms as nodes and bonds as weighted edges for predictive modeling.
- Pattern Recognition: Implemented a high-performance substructure search method utilizing molecular fingerprints to detect specific chemical motifs within large-scale datasets.

Core Skills

Analytical Instrumentation & Methods: HPLC, LC-MS, GC, ICP-MS, TLC, gamma spectroscopy, endotoxin testing, method development, method validation

Data & Programming: Python, SAS, SQL, R, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow

Laboratory Systems & Compliance: Laboratory Information Management Systems (LIMS), Equipment IQ/OQ/PQ, GMP, FDA/ICH guidelines, QMS documentation

Software & Tools: Flask, Streamlit, React.js, Next.js, Docker, Git, Microsoft Office

version control

Education

University of California - Berkeley
May 2024 - May 2026

Master's Molecular Science and Software Engineering
GPA: 3.9

Michigan State University
Sep 2015 - May 2019

Bachelor's Chemistry
GPA: 3.85